TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 12/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Huỳnh Công Việt Ngữ		nguhcv@fpt.edu.vn	

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Nguyễn Minh Trí	SE192645	0785339955	chinjsu130205@gmail.com	Leader
2	Đỗ Quốc Bảo	SE193803	0965928958	doquocbao2009@gmail.com	Member
3	Nguyễn Công Thành	SE196309	0353028372	ngocthanhkt126@gmail.com	Member
4	Võ Đồng Đức Khải	SE196405	0983529414	khaiqn2707@gmail.com	Member
5	Trần Quốc Dinh	SE196495	0868495138	nguyensoi0966622100@gmail.com	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***HistoryTalk: A SaaS Historical Education Platform Powered by Conversational AI and RAG Technology***

3.1.2. Vietnamese: ***HistoryTalk: Nền tảng SaaS Giáo dục Lịch sử ứng dụng Trí tuệ Nhân tạo và công nghệ RAG***

(*) 3.2. Main proposal content (including result and product)

a) Context:

Currently, students often face difficulties and feel bored with History due to traditional approach methods that favor rote learning (cramming dates and events). The lack of context and interaction leads to quick forgetting. Besides, schools and teachers lack specialized EdTech tools to assign lessons, manage classes, and track students' self-study progress automatically.

The project was born to solve the above problem by building HistoryTalk - a Software as a Service (SaaS) platform, providing interactive learning tools for students (B2C) and a classroom management system dedicated to the school block (B2B).

b) Proposed Solutions:

Build the SaaS HistoryTalk platform, operating two models simultaneously: B2C (for individual users) and B2B (for schools). Specifically, the solution includes:

- Conversational AI & RAG: Utilizing LLMs combined with RAG technology, students can engage in direct conversations with historical figures. AI data is verified against standard databases to minimize hallucinations.
- Visualization & Assessment: Lesson videos integrated with historical maps and a quiz system to reinforce knowledge.
- SaaS Management (School Management): Provides a dedicated module for Teachers and Schools to manage student lists, assign tasks/tests (context, characters, quizzes/tests), and monitor learning progress.

Example of the Learning Journey on the system:

Step 1: Read the core summary of the historical event Context and watch the lesson video.

Step 2: Chat with AI playing the role of a Historical Character to clearly understand the causes, motivations, and questions of each individual.

Step 3: View the geographical location where the event took place on the Map, explore battles and locations in different timelines.

Step 4: Do Quizzes / Tests to check knowledge.

c) Functional Requirements:

Module 1 — Authentication & Authorization

- Register/Login via Email & Password, integrated Google Login.
- Individual Flow: Handle Customer account upgrades to Premium packages.
- SaaS Flow: System Admin grants School Admin accounts. From there, School Admin creates Teacher and Student accounts.

Module 2 — Classroom Management (SaaS)

- School Admin: Create, edit, delete Classrooms; Manage the school-wide Classroom list.
- **Teacher:**
  - Add/delete students in the classes they are in charge of.
  - Assign: Designate students to learn Events (Context), interact with Characters (Character), and do Quizzes.
  - Create local Content (only visible to students in class): Draft Quizzes/Tests, Events, and Characters according to local history.

Module 3 — Learning & Interaction

- Display a list of Events, Historical Characters, and Lesson Videos.
- Interact with AI playing a Character (Support: Text chat, Voice messages).
- View and interact with an Interactive Map.
- Do practice Quizzes and take Tests for grades.

Module 4 — Token Management
- Manage daily Token balances for Customers and School Students.
- Block Chat features when out of Tokens, displaying suggestions for upgrade packages.
- Accumulate new Tokens daily when users log in.

Module 5 — Progress Tracking & Analytics Reports
- Customer and School Student: View Quiz history and progress completion of assigned tasks.
- Teacher: Track students' Quiz scores in class; View interaction statistics (number of dialogue messages with AI) for each student.

Module 6 — Payment (Individual)
- Manage Premium service packages.
- Integrate online payment gateways.
- Store transaction history and package status.

Module 7 — System Content Administration (Content Admin)
- Manage the general academic database (Global): Events, Characters, Documents, Quizzes.
- Manage the display status (Publish / Inactivate) of content.

Module 8 — Mobile App
- Development roadmap for Mobile application to optimize the learning experience anytime, anywhere for Customers and School Students.

d) Non-Functional Requirements:
- Performance: Chat API returns in real-time (Streaming Response) to ensure smoothness.
- Accuracy: RAG (Retrieval-Augmented Generation) architecture retrieves context from Vector DB to force AI to answer according to historical data, minimizing Hallucinations.
- Security & Integrity: Multi-tenant architecture securely separates data between Schools (SaaS). Security for payment APIs.
- Usability: Modern UI/UX, easy to use for all users.

e) Theory & Practical:
- Theoretical Basis: Learning through interactive and personalized experiences (Interactive Learning and Personalized Learning).
- **Core Technology:**
  - AI: Python processing Chatbox hosting, Session Customize, Prompt Engineering, RAG Pipeline system.
  - Backend: Java (Spring Boot) processing Core & Logic; Node.js processing Real-time/Proxy.
  - Frontend: Next.js / React / Tailwind CSS.
  - Database: PostgreSQL, Vector Database.

f) Products (Expected Deliverables):

1. SaaS Platform: Live running HistoryTalk software system
2. Source Code: Frontend, Backend (Java & Node.js), AI Proxy source code.
3. Technical Documentation: System Architecture, Database Schema, API Documentation.

g) Proposed Tasks:

- Nguyễn Minh Trí (Project Leader & AI Engineer & Backend Developer): Manage general project progress (Agile/Sprints), survey business flows. Responsible for the core AI and BE chat: Building and refining RAG, handling Streaming AI Chat flows, optimizing Prompts to ensure historical accuracy, managing Tokens.
- Nguyễn Công Thành (Frontend Developer & Tester): Responsible for the entire interface system: Web & App programming for students (Chat, Map, Quiz) and B2B SaaS management portals (Teacher/Admin) and Internal Management (Admin Content, Admin System). Handle UI/UX testing.
- Võ Đồng Đức Khải (Backend Developer & Tester): Design Multi-tenant Database. Develop Core APIs for Authentication and Authorization (Role-based auth), OAuth with Google, participate in developing User Management APIs, Classroom Management APIs, build Content Management APIs (CRUD Content, Character, Document), module for managing document uploads like PDFs as Knowledge base. Handle unit & integration testing.
- Trần Quốc Dinh (Backend Developer & DevOps): Responsible for B2B SaaS sub-system: participate in developing Class Management and User Management APIs, learning progress tracking system (Progress Tracking & Analytics) and system administration (System Tracking & Analytics). Deploy BE + AI Server integrated with CI/CD automation.
- Đỗ Quốc Bảo (Backend Developer & Tester): Build APIs for Quiz flow and management, handle Online Payment system logic, and interactive map APIs. Handle test case design and execution.

4. Other comments (propose all relative things if have):

.....
.....
.....

HCM, date 05/09/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

Registration Form – Record of Changes

Form BM03. Attach to the proposed final registration form. The approved project code and title remain unchanged. Export to PDF together with the registration form. File name: SWP490_<ProjectCode>_R2_Registration.pdf

1. Project Information

Project code	FA26SE295
Project name	HistoryTalk: A SaaS Historical Education Platform Powered by Conversational AI and RAG Technology (HistoryTalk: Nền tảng SaaS Giáo dục Lịch sử ứng dụng Trí tuệ Nhân tạo và công nghệ RAG)
Team	Nguyễn Minh Trí (SE192645 - Leader), Đỗ Quốc Bảo (SE193803), Nguyễn Công Thành (SE196309), Võ Đồng Đức Khải (SE196405), Trần Quốc Dinh (SE196495)
Supervisor	Huỳnh Công Việt Ngữ (nguhcv@fpt.edu.vn)
Current registration version	v1.0 (05/09/2026)
Proposed version	v2.0 (23/10/2026)
Version	1.1
Date	23/10/2026

2. Changes

☐ No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	General Information (Duration time)	Duration time: from 09/2026 To 03/2027	Duration time: from 09/2026 To 12/2027	Correct typographical error in the project end date to accurately reflect the official Fall 2026 semester timeline.	Nguyễn Minh Trí (Leader)
2	2. Register information for students (No. 1 & No. 3)	No. 1: Nguyễn Công Thành | SE192645 | 0785339955 | chinjsu130205@gmail.com | Role: [Blank] No. 3: Nguyễn Minh Trí | SE183033 | 0938132218 | trinmse183033@fpt.edu.vn | Role: Leader	No. 1: Nguyễn Minh Trí | SE192645 | 0785339955 | chinjsu130205@gmail.com | Role: Leader No. 3: Nguyễn Công Thành | SE196309 | 0353028372 | ngocthanhkt126@gmail.com | Role: Member	Correct typographical and data entry errors in student codes, phone numbers, email addresses, and group roles from initial registration.	Nguyễn Minh Trí (Leader)
3	3.2.g. Proposed Tasks (Nguyễn Công Thành)	...Participate in writing Test Cases and system testing.	...Handle UI/UX testing.	Refine and specialize testing responsibilities based on Iteration 1 findings to align directly with Frontend development tasks.	Nguyễn Minh Trí (Leader)
4	3.2.g. Proposed Tasks (Võ Đồng Đức Khải)	...Participate in writing Test Cases and system testing.	...Handle unit & integration testing.	Refine and specialize testing responsibilities based on Iteration 1 findings to ensure	Nguyễn Minh Trí (Leader)

No.	Section	Current content	Proposed content	Reason	Proposed by
				Backend core API and multi-tenant database reliability.	
5	3.2.g. Proposed Tasks (Đỗ Quốc Bảo)	...Participate in writing Test Cases and system testing.	...Handle test case design and execution.	Clarify specific QA ownership for test case design and end-to-end execution across functional modules.	Nguyễn Minh Trí (Leader)

Accepted reasons: Review 1 feedback, findings from Iteration 1, or constraints from data, partners or technology. Changes that significantly reduce business complexity must be fully justified.

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
1		
2		
3		
4		
5		